

Vendor Management Inventory Analysis Project

By: Abdullah Qatu

Introduction	2
Business Context	2
Problem Statement	2
Stakeholders	2
Approach	2
Data Acquisition & Exploration	3
Data Requirements and Sources	3
Data Cleansing and Formatting	4
Data Exploration	4
Transformations	4
Analysis & Visualizations Selection	6
Project Result	7
Benefits	7
Challenges & Learnings	7
Challenges	7
Learnings	9
Conclusion and Impact	9
Skills Showcased	9
Impact and follow up	10

Table of Abbreviations

Abbreviation	Meaning
VMI	Vendor Managed Inventory
SAP BW	SAP Business Warehouse
DSC	Distributor Stock Cover
SISO	Sell In/ Sell out
MDX	Multidimensional Expression
ARS	Automatic Replenishment System
ADF	Azure Data Factory
IMS	Inter-Market Supply

Introduction

Business Context

The Vendor Managed Inventory (VMI) team is responsible for the management of the stock cover for each of Nestle's distributors, ensuring optimal inventory replenishment by analyzing delivery schedules, stock demand, and the sell in and sell out of the products.

By leveraging this data, the team does not only enhance the efficiency of the supply chain's processes but also support Nestlé's commitment to delivering high-quality products to consumers in a timely manner.

Problem Statement

The teams' main struggle was the time consumed to retrieve the data needed to fulfill their work from multiple data sources. The team also had to parse, format, aggregate, and analyze the information in order to obtain the results they needed in order to make their informed decisions.

Stakeholders

The main stakeholders were the vendor managed inventory team across 7 countries (Jordan, KSA, Oman, Qatar, Kuwait, Lebanon, Iraq) and their team lead. They were the end users of this report.

Approach

After multiple meetings and discussions with the VMI team, I was able to capture the business nuances needed to fully understand their line of work and identify what were the essential factors that contributed to their success which are having accurate stock measures, accurate stock forecasts, and the effective communication and engagement with the distributors.

With that said, I deemed the optimal approach was to create a Power Bi report that consolidated all the data the team needed with addition to the transformations required in order to have full visibility of the stock and have the ability to streamline their decision-making process. The details of the approach will be outlined later in this document.

Data Acquisition & Exploration

Data Requirements and Sources

This section will outline the main data requirements for the project as well as their sources which go as follows:

- **Distributor Stock Cover (DSC):**

Description: The DSC covers the stock cover for each SKU per distributor with all related information such as stock date, stock quantity, and stock value.

Source: The DSC data was retrieved through the use of an Multidimensional Expression (MDX) statement through the use of the SAP Business Warehouse connector on Power Bi.

- **Sell In/Sell Out (SISO):**

- **Description:** The SISO covers the sell in and sell out quantities and values for each SKU per distributor.

- **Source:** The SISO data was also retrieved through the use of an Multidimensional Expression (MDX) statement through the use of the SAP BW connector on Power Bi.

- **Automatic Replenishment System (ARS):**

- **Description:** The ARS file contains all the information related to the proposals the VMI team sends out to the distributors. The information contains the proposal number, the SKUs included, the proposal's quantities, amounts, stock forecasts, and more.

- **Source:** The ARS file is added manually to a cloud storage account where a pipeline on the Azure Data Factory ingests the file into an SQL database that is connected to the Power Bi.

- **Inter-Market Supply (IMS):**

- **Description:** The IMS file contains the stock target of each SKU for each distributor for a specific month.
- **Source:** The IMS file is also added manually to a cloud storage account where a pipeline on the Azure Data Factory ingests the file into an SQL database that is connected to the Power Bi.

Data Cleansing and Formatting

After observing the files in Power Bi, I discovered numerous columns in the Distributor Stock table were irrelevant to the needs of the team as well as multiple columns in the SISO table containing more than 90% missing records. The approach used was removing these columns.

Some formatting of the data types was needed such as parsing certain date fields to become mdy format in order for the Power Bi to render it as a date. Furthermore, splitting certain columns was required to extract the distributor code (Ex. 4465119-GSK to 4465119)

Data Exploration

Data exploration was conducted with the VMI team on Power Bi through plugging in the data in certain visuals to make sense of the data and ensure the overall distribution was as expected. Tests were run by comparing the report's data with the old system to ensure the data is accurate and complete.

Data Analysis & Visualization

Transformations

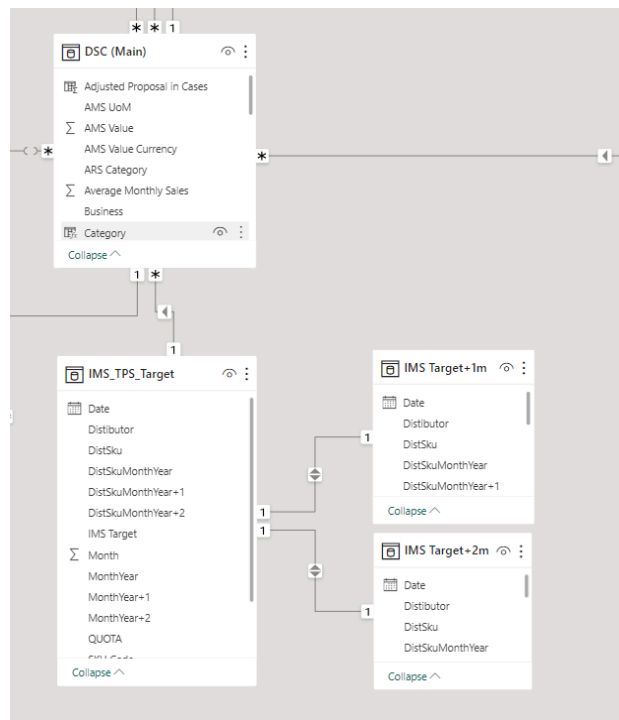
This section will explain each analysis requirement with the transformation used to achieve the result:

- **Establishing Entity Relationships:**

- **Requirement:** Establishing the relationships between all the entities in the data model.
- **Transformation:** A composite key was created and used to join all the tables in the model. The key consists of the distributor, SKU, and the date or the month and year depending on the entity.

- **Consecutive IMS Targets on same row:**

- **Requirement:** To visualize in the DSC table the IMS target for each SKU, the IMS target after 1 month and after 2 months all on the same record.
- **Transformation:** 2 Temp tables were created based on calculating the offset of the IMS target table by 1 and 2 months respectively. After that, the 2 tables are linked back to the original IMS table and a lookup function is used in the DSC table to retrieve the 3 IMS targets.



The relationship of the offset tables.

- **Next Proposal Date:**

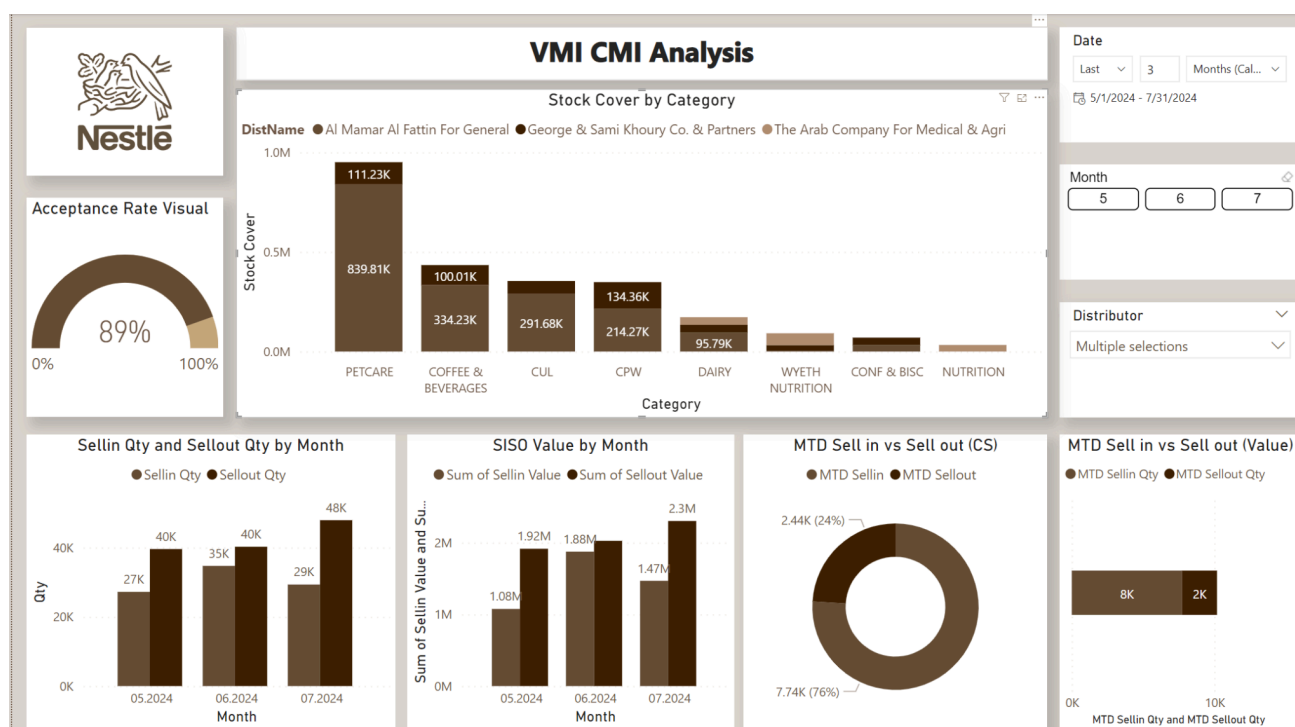
- **Requirement:** To visualize in the ARS proposal table for each SKU what the next proposal date will be.
- **Transformation:** A measure was made to calculate the minimum date in the proposal dates table taking into account the selected distributor and date range in the visualization pane in order to retrieve the next proposal date.

Analysis & Visualizations Selection

The data was primarily segmented by the distributor and the proposals and time series analysis was used to analyze the data. The stakeholder wanted to see the change of the stock over periods of time such as MTD, daily, and on a monthly basis.

The analysis and choice of visualizations were determined based on the data provided and with the feedback of the stakeholders in terms of what is their preferred method of visualizing the data.

Stacked column charts were the preferred method of visualizing the stock cover across categories and distributors while the pie charts and clustered column charts were chosen in visualizing the SISO data on a month by month and MTD time periods.



A tabular view was also used by the team to analyze proposals to observe all the data pertaining to each SKU in the proposal. For example, the table shows the stock cover, SISO value, quantity, next proposal date, current IMS target, IMS target after 1 month, and IMS target after 2 months all on the same record.

Project Result

Initial feedback of the report was very positive, the stakeholders were pleased with the outcome of the report and the team's main pain points were resolved.

Benefits

- 50 man hours per week have been saved.
- Errors taking place during extraction/parsing/formatting have been eliminated.
- The report gives a standardized view of the data that is shared across the region. This improves the communication and efficiency between team members.
- The data model used in the report can be reused for other use cases in the company.

Challenges & Learnings

Challenges

The main problems and their solutions that came up during the project go as follows:

- Unclear Requirements:
 - **Problem:** The stakeholders were not sure what the requirements were, the meetings dragged on more than they should as the stakeholders took the time discovering what they might or might not want.
 - **Solution:** I used a set of very specific questions that aimed to guide the stakeholder to what fulfills their project needs while still remaining technically feasible.

- **Complex Calculations**
 - **Problem:** Certain formulas such as the IMS targets and creating the logical link between the tables were challenging.
 - **Solution:** I came up with the solutions by reading documentations, forums, and the process of brainstorming with colleagues and business domain experts in order to come up with the solution.
- **Project Scheduling Issues and Urgency:**
 - **Problem:** Certain team members have placed the project in the low priority list leading to slow response times and the team had no sense of urgency in completing the project.
 - **Solution:** I communicated and motivated the team about completing the project by emphasizing the benefits and the recognition they will receive when the project is completed.
- **Unfamiliarity With MDX Statements:**
 - **Problem:** I was unfamiliar with MDX queries and their structure, formulating the correct statement was a challenge.
 - **Solution:** Going through the MDX documentation and observing how other existing MDX statements were used helped me understand the structure of the statement and how to effectively use it.
- **Establishing Connection To Cloud Storage:**
 - **Problem:** Establishing a direct connection to a CSV file on the cloud storage through Power Bi posed a challenge because whenever the file was changed, the connection to the file breaks.
 - **Solution:** The file was ingested to an SQL database through an ADF pipeline where the connection was made to the database instead thus having a seamless connection to the file.

Learnings

The main things I learned during the project:

- I have gained business knowledge regarding the operations of the SCM function. This will help me manage more SCM projects in the future.
- I gained a thorough understanding of how MDX queries work as well as how to apply proper logic when creating relationships between tables.
- I learned to ensure the requirements are well accounted for by all parties before the execution of the project in order to reduce wasted time and rework occurring at later stages of the project cycle.
- I learned how to motivate a team to successfully complete a project and show a sense of initiative in their work.

Conclusion and Impact

The VMI analysis project was a great opportunity for me to learn and to showcase my skills as a data analyst where I can see my skills provide value to not just myself but for the colleagues I work with.

Skills Showcased

- Demonstrated project management skills by independently overseeing the entire project from start to finish.
- Showcased my analytical skills by planning and executing the entire data extraction, reporting, and visualization pipeline.
- Exhibited interpersonal skills by effectively communicating and motivating the VMI team, resulting in improved efficiency and effectiveness in their work.

Impact and follow up

After successfully completing the project, the VMI team across 7 countries will benefit from using the report in their day to day tasks. They can effectively and efficiently manage their distributors' stock cover as well as be able to analyze the trends and forecasts of their stock. As a result, the VMI workload has been significantly reduced by 50 man hours per week thus allowing them to allocate more time and resources towards enhancing other areas where they can deliver even greater value.

The team will continue working on the report until the next phase of creating a multi-country report dedicated to the team lead in order to analyze the inventory data across multiple countries.